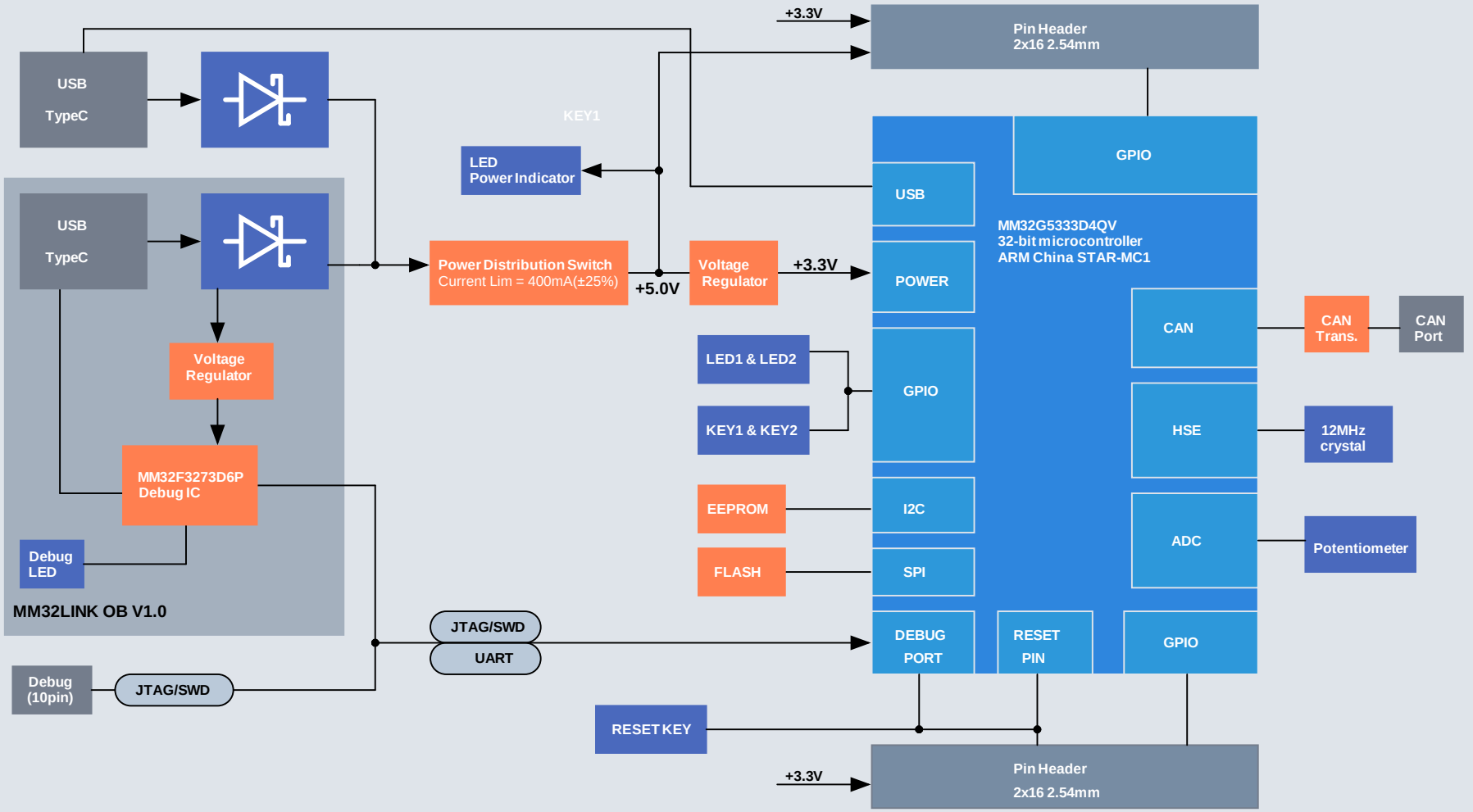
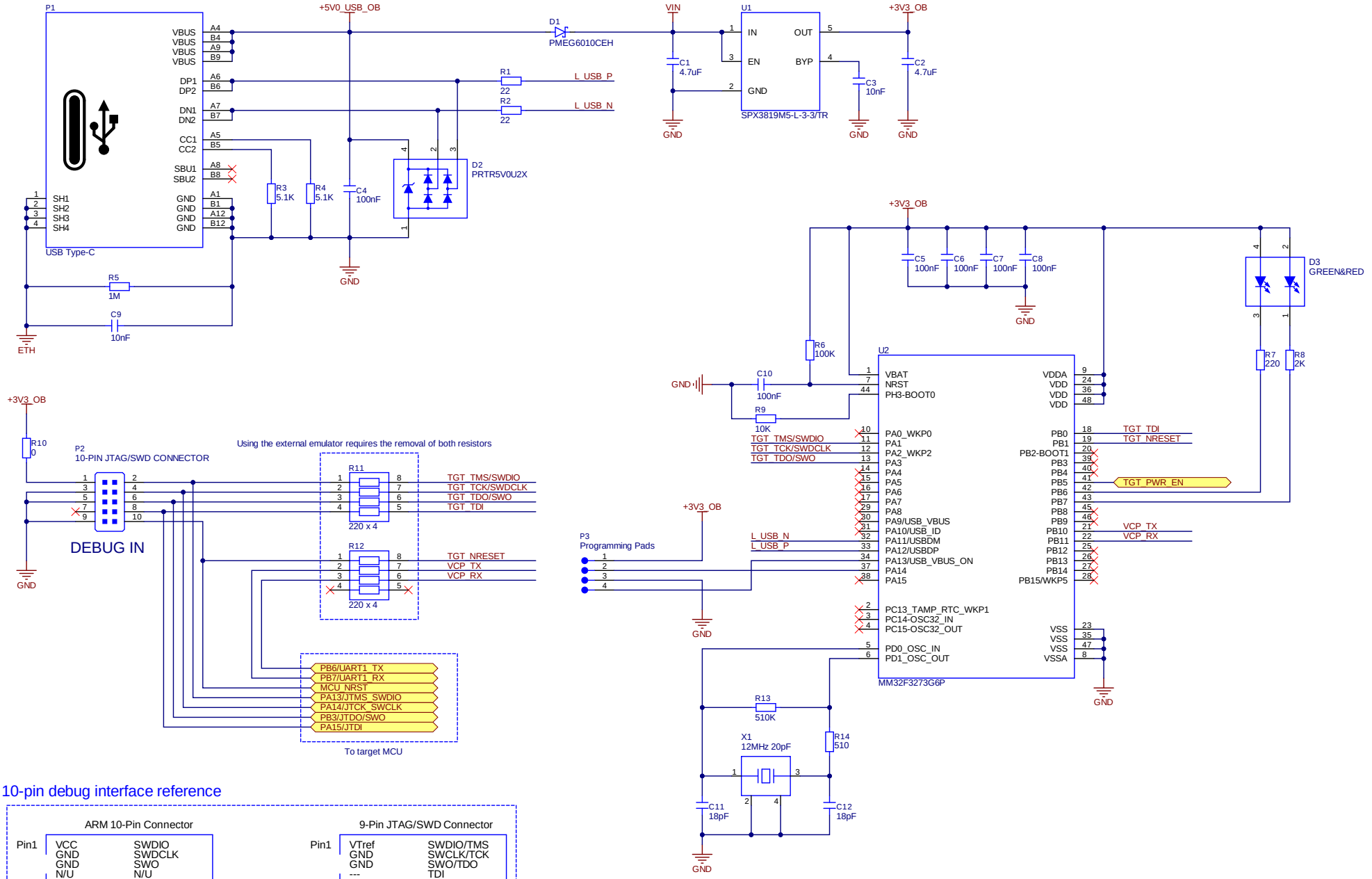
Mini-G5333-OB



Connector | General Component | Chip | Target MCU

Project Title	MB101-Mini-G5333-OB		
Sheet Title	01---System Block Diagram		
Size A4	Rev A	Drawn	Lu Weiwei
Sheet 1 / 9		Checked	Li Hongpeng
Data 2023.07.03		Approved	Chen Do

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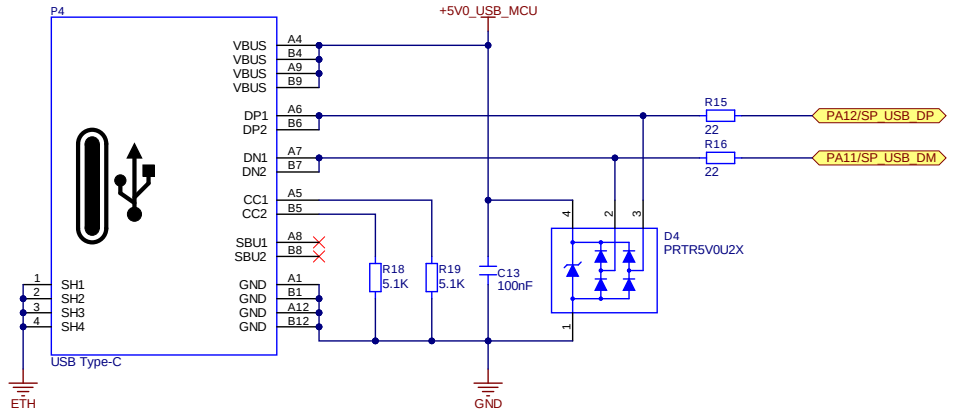
10-pin debug interface reference

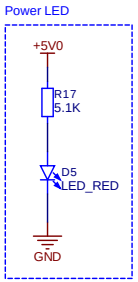
ARM 10-Pin Connector			9-Pin JTAG/SWD Connector		
Pin1	VCC	SWDIO	Pin1	VTref	SWDIO/TMS
	GND	SWDCLK		GND	SWCLK/TCK
	GND	SWO		GND	SWO/TDO
	N/U	N/U		---	TDI
	GND	RESET		NC	nRESET
	1.27mm/50mil			1.27mm/50mil	

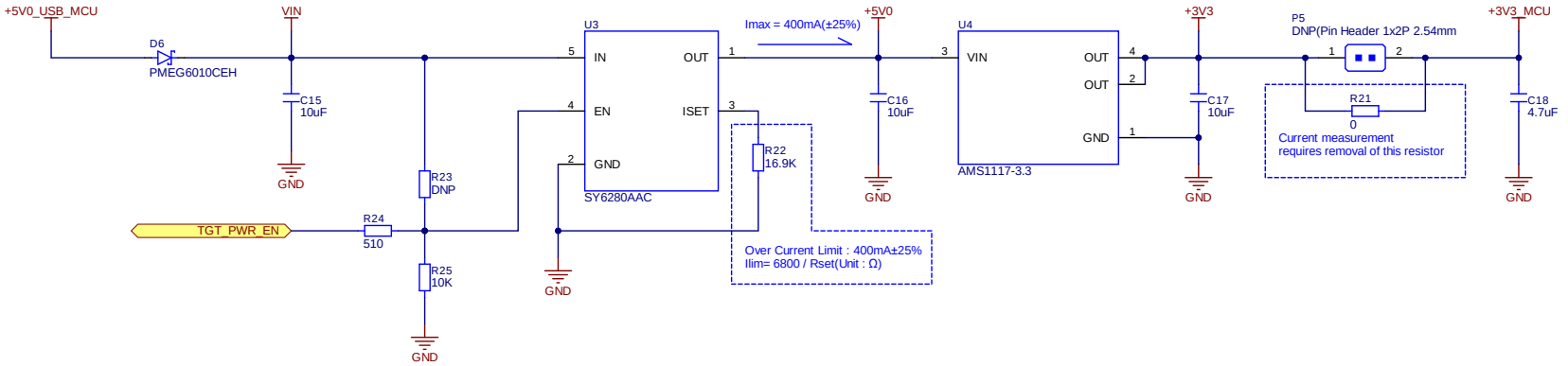
DNP : DO NOT PLACE

Project Title	MB101-Mini-G5333-OB		
Sheet Title	02---MM32Link-OB V1.0		
Size A3	Rev A	Drawn	Lu Weiwei
Sheet 2 / 9		Checked	Li Hongpeng
Data 2023.07.03		Approved	Chen Do

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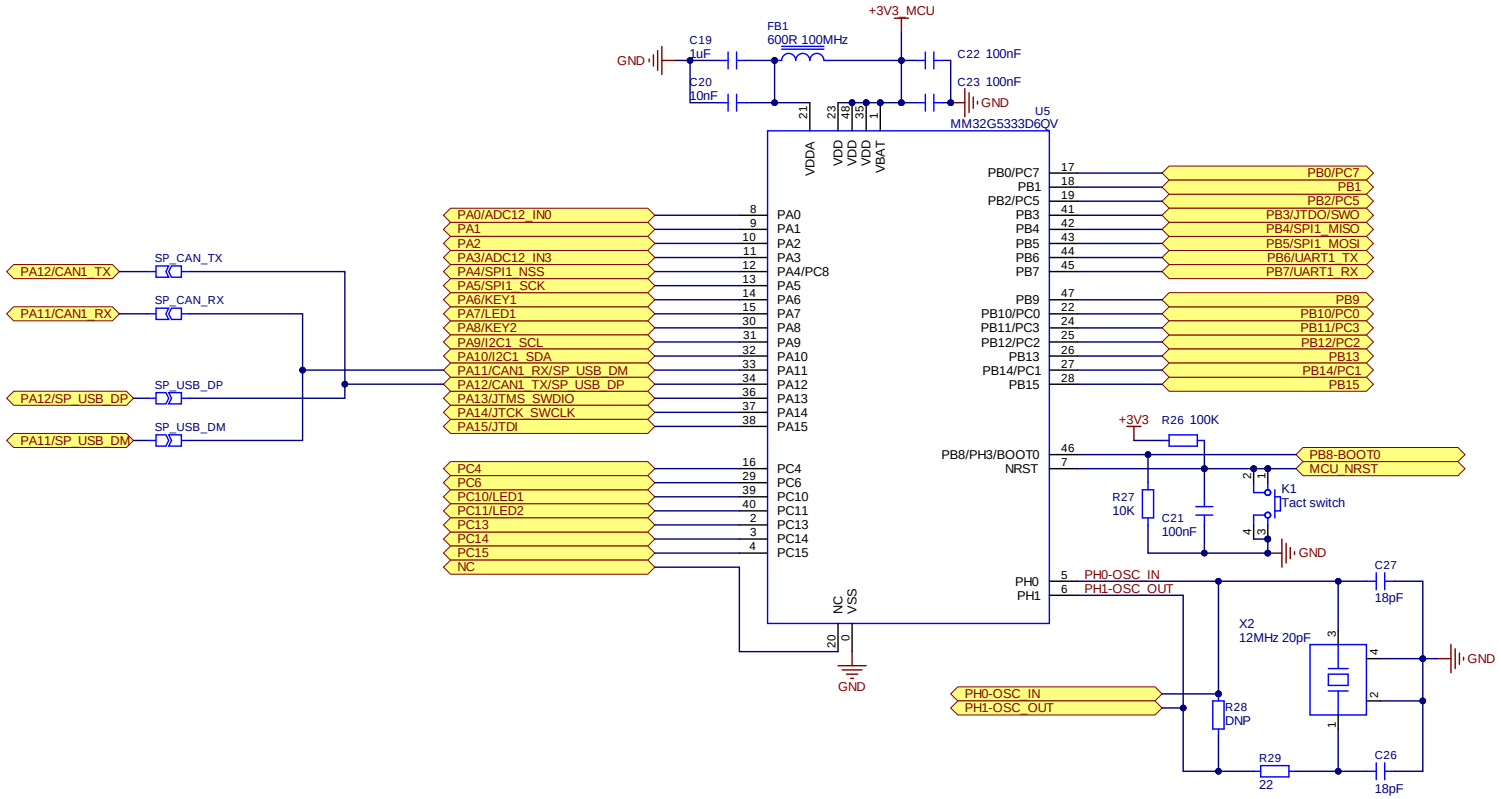




DNP : DO NOT PLACE

Project Title	MB101-Mini-G5333-OB				
Sheet Title	03---Target Power				
Size	A3	Rev	A	Drawn	Lu Weiwei
Sheet	3 / 9			Checked	Li Hongpeng
Data	2023.07.03			Approved	Chen Do

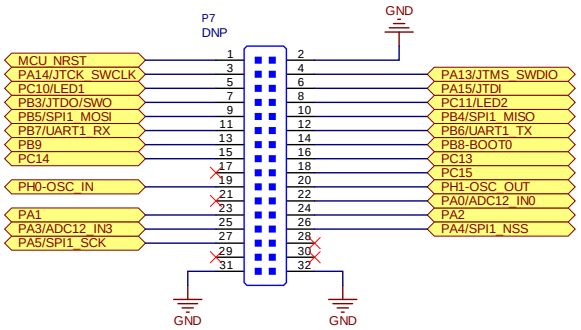
MindMotion SoC Solutions



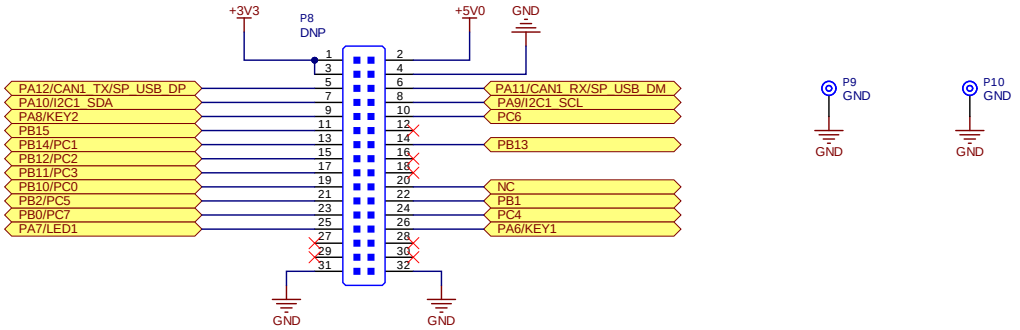
Project Title	MB101-Mini-G5333-OB		
Sheet Title	04---Target MCU		
Size A3	Rev A	Drawn	Lu Weiwei
Sheet 4 / 9		Checked	Li Hongpeng
Data 2023.07.03		Approved	Chen Do

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Left Part:



Right Part:



Pin assignment:

PIN08---PA0 / TIM2_CH1 / TIM2_ETR / TIM5_CH1 / TIM8_ETR / UART2_CTS / UART4_TX / I3C_SCL / ADC12_IN0 / COMP12_INP0 / COMP1_INM2 / WKP0
PIN09---PA1 / TIM2_CH2 / TIM5_CH2 / UART2_RTS / UART4_RX / I3C_SDA / ADC12_IN1 / COMP12_INP1
PIN10---PA2 / TIM2_CH3 / TIM5_CH3 / UART2_TX / COMP2_OUT / LPUART_TX / ADC12_IN2 / COMP12_INP2 / COMP2_INM2 / WKP2
PIN11---PA3 / TIM2_CH4 / TIM5_CH4 / I2S1_MCK / UART2_RX / LPUART_RX / ADC12_IN3 / COMP12_INP3
PIN12---PA4 / TIM5_ETR / SPI1_NSS / I2S1_WS / SPI3_NSS / I2S3_WS / USART_TX / ADC12_IN4 / COMP12_INM0 / DAC1_OUT
PIN12---PC8---TIM3_CH3---I2C2_SCL
PIN13---PA5---TIM2_CH1---TIM2_ETR---TIM3_ETR---TIM8_CH1N---SPI1_SCK---I2S1_CK---USART_RX---ADC12_IN5---COMP12_INM1
PIN14---PA6---TIM1_BKIN1---TIM3_CH1---TIM8_BKIN1---SPI1_MISO---I2S1_MCK(extSD)---COMP1_OUT---ADC12_IN6
PIN15---PA7---TIM1_CH1N---TIM3_CH2---TIM8_CH1N---SPI1_MOSI---I2S1_SD---COMP2_OUT---CRS_SYNC---MDSW_IN1---ADC12_IN7
PIN30---PA8 / MCO1 / TIM1_CH1 / TIM3_ETR / I2C1_SCL / USART_CK
PIN31---PA9 / TIM1_CH2 / I2C1_SCL / UART1_TX / USB_FS_VBUS / USART_TX
PIN32---PA10 / TIM1_CH3 / I2C1_SDA / UART1_RX / USB_FS_ID / USRAT_RX
PIN33---PA11 / TIM1_CH4 / I2C1_SMBA / UART1_CTS / COMP1_OUT / CAN1_RX / USART_CTS / USB_FS_DM
PIN34---PA12 / TIM1_ETR / UART1_RTS / COMP2_OUT / CAN1_TX / USART_RTS / USB_FS_DP
PIN36---PA13 / JTMS_SWDIO / I2C1_SMBA / USB_FS_VBUSON / MDSW_OUT2
PIN37---PA14 / JTCK_SWCLK / I2C1_SDA / I2S3_MCK / UART4_CTS / MDSW_IN4
PIN38---PA15 / JTDI / TIM2_CH1 / TIM2_ETR / I2C1_SCL / SPI1_NSS / I2S1_WS / SPI3_NSS / I2S3_WS / UART4_RTS

PIN17---PB0 / TIM1_CH2N / TIM3_CH3 / TIM8_CH2N / ADC12_IN9
PIN17---PC7 / TIM3_CH2 / TIM8_CH2 / I2C1_SDA / SPI3_MISO / I2S3_MCK(extSD) / SPI3_MISO / I2S3_MCK(extSD)
PIN18---PB1 / TIM1_CH3N / TIM3_CH4 / TIM8_CH3N / ADC12_IN9
PIN19---PB2 / TIM1_CH4N / TIM8_CH4N / BOOT1
PIN19---PC5 / UART1_RX / USART_RX / ADC2_IN15 / WKP3
PIN41---PB3 / JTDO / TRACESWO / TIM2_CH2 / TIM4_ETR / SPI1_SCK / I2S1_CK / SPI3_SCK / I2S3_CK / I3C_SCL / USART_CTS
PIN42---PB4 / NJTRST / TIM3_CH1 / SPI1_MISO / I2S1_MCK(extSD) / SPI3_MISO / I2S3_MCK(extSD) / I3C_SDA / USART_RTS
PIN43---PB5 / TIM3_CH2 / TIM8_CH3N / I2C1_SMBA / SPI1_MOSI / I2S1_SD / SPI3_MOSI / I2S3_SD / LPTIM_CH1 / WKP4
PIN44---PB6 / TIM4_CH1 / TIM8_CH4N / I2C1_SCL / SPI1_NSS / I2S1_WS / SPI3_NSS / I2S3_WS / UART1_TX / LPTIM_ETR / USART_TX
PIN45---PB7 / TIM4_CH2 / I2C1_SDA / I2S1_MCK / I2S3_MCK / UART1_RX / USART_RX / MDSW_OUT3
PIN46---PB8 / TIM4_CH3 / TIM8_CH1N / I2C1_SCL / UART1_CTS / COMP1_OUT / CAN1_RX / USART_CTS
PIN46---PH3 / BOOT0
PIN47---PB9 / TIM4_CH4 / TIM8_CH2N / I2C1_SDA / SPI2_NSS / I2S2_WS / UART1_RTS / COMP2_OUT / CAN1_TX / USART_RTS
PIN22---PB10 / TIM2_CH3 / I2C2_SCL / SPI2_SCK / I2S2_CK / UART3_TX / LPTIM_CH1 / LPUART_RX
PIN22---PC0 / TIM1_BKIN3 / I2C1_SCL / UART4_TX / LPUART_RX / ADC12_IN11
PIN24---PB11 / TIM2_CH4 / I2C2_SDA / I2S2_MCK / UART3_RX / LPTIM_ETR / LPUART_TX / MDSW_OUT1
PIN24---PC3 / I2C2_SDA / SPI2_MOSI / I2S2_SD / UART4_RTS / ADC1_IN13
PIN25---PB12 / TIM1_BKIN2 / TIM5_ETR / I2C2_SMBA / SPI2_NSS / I2S2_WS
PIN25---PC2 / I2C2_SCL / SPI2_MISO / I2S2_MCK(extSD) / UART4_CTS / ADC1_IN12

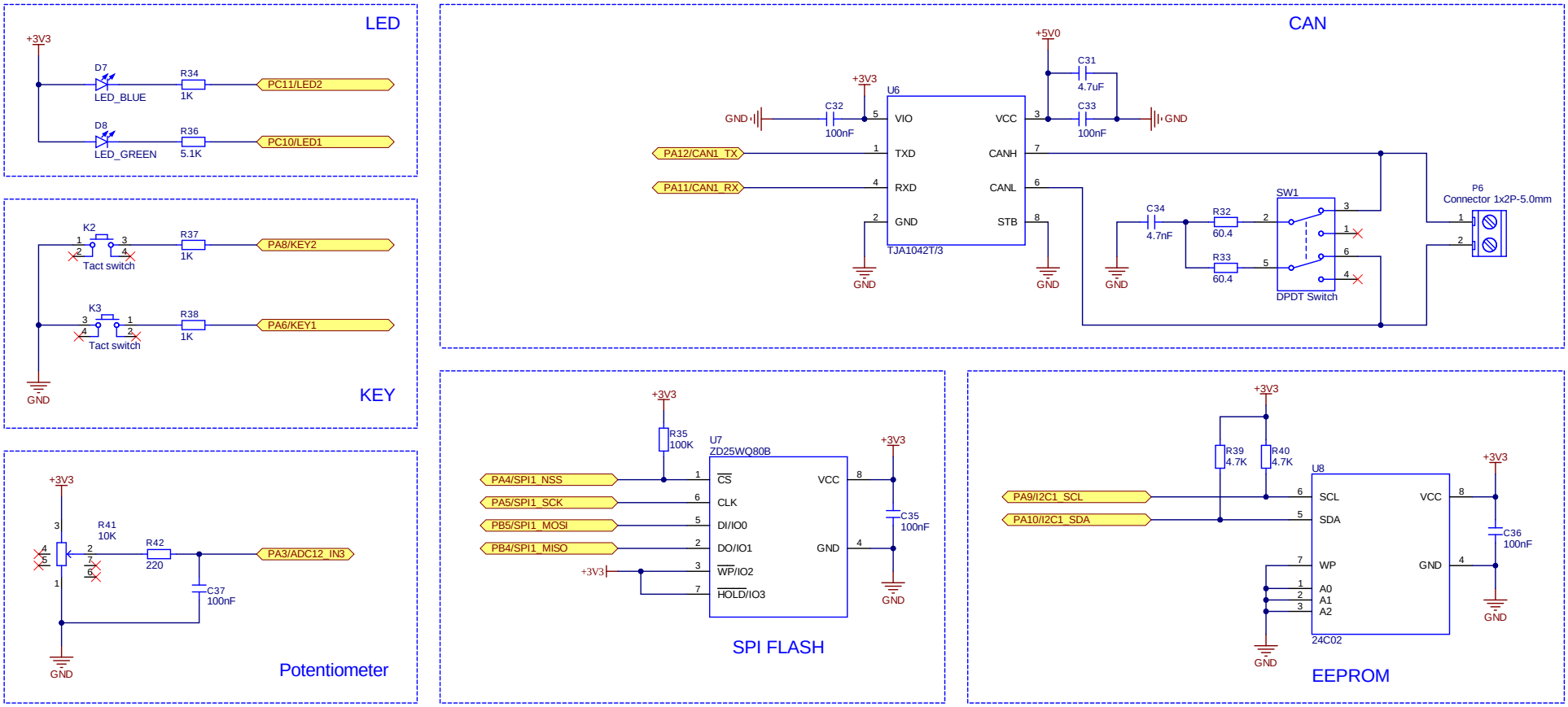
PIN26---PB13 / TIM1_CH1N / TIM8_CH1N / SPI2_SCK / I2S2_CK / UART3_CTS
PIN27---PB14 / TIM1_CH2N / TIM8_CH2N / SPI2_MISO / I2S2_MCK(extSD) / UART3_RTS / MDSW_IN2
PIN27---PC1 / TIM8_BKIN2 / I2C1_SDA / UART4_RX / LPUART_TX / ADC12_IN11
PIN28---PB15 / TIM1_CH3N / TIM8_CH3N / SPI2_MOSI / I2S2_SD / WKP5

PIN16---PC4 / UART1_TX / USART_TX / ADC2_IN14
PIN29---PC6 / TIM3_CH1 / TIM8_CH1 / I2C1_SCL / SPI2_MISO / I2S2_MCK(extSD)
PIN39---PC10 / SPI3_SCK / I2S3_CK / UART3_TX / UART4_TX / I3C_SCL
PIN40---PC11 / SPI3_MISO / I2S3_MCK(extSD) / UART3_RX / UART4_RX / I3C_SDA
PIN02---PC13 / TIM8_CH1 / MDSW_IN0 / WKP1 / BKP_TAMPER
PIN03---PC14 / TIM8_CH2 / OSC32_IN
PIN04---PC15 / TIM8_CH3 / OSC32_OUT

PIN05---PH0 / OSC_IN
PIN06---PH1 / OSC_OUT

Project Title	MB101-Mini-G5333-OB		
Sheet Title	05---Target MCU IO Expansion		
Size	A3	Rev A	Drawn Lu Weiwei
Sheet	5 / 9		Checked Li Hongpeng
Data	2023.07.03		Approved Chen Do

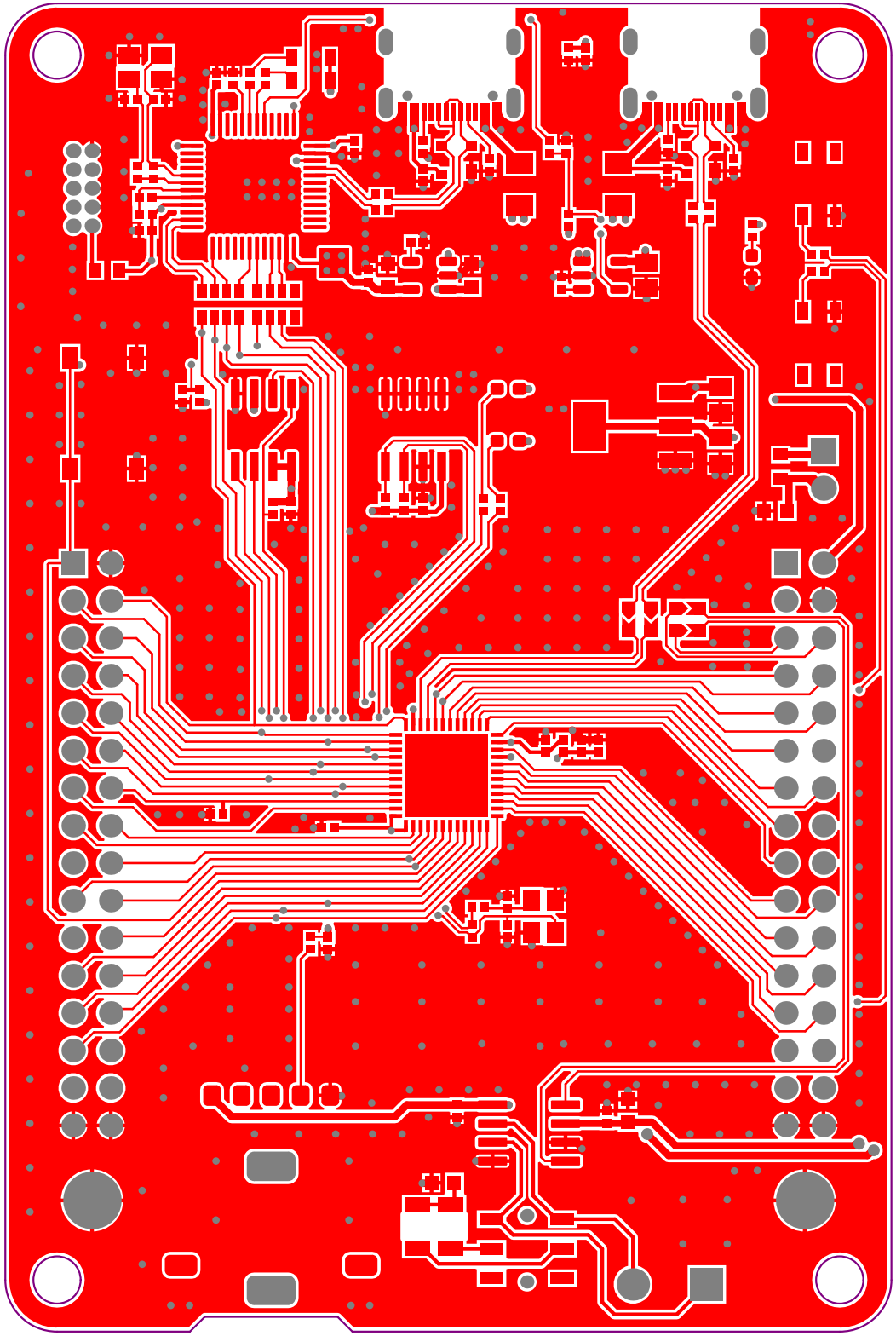
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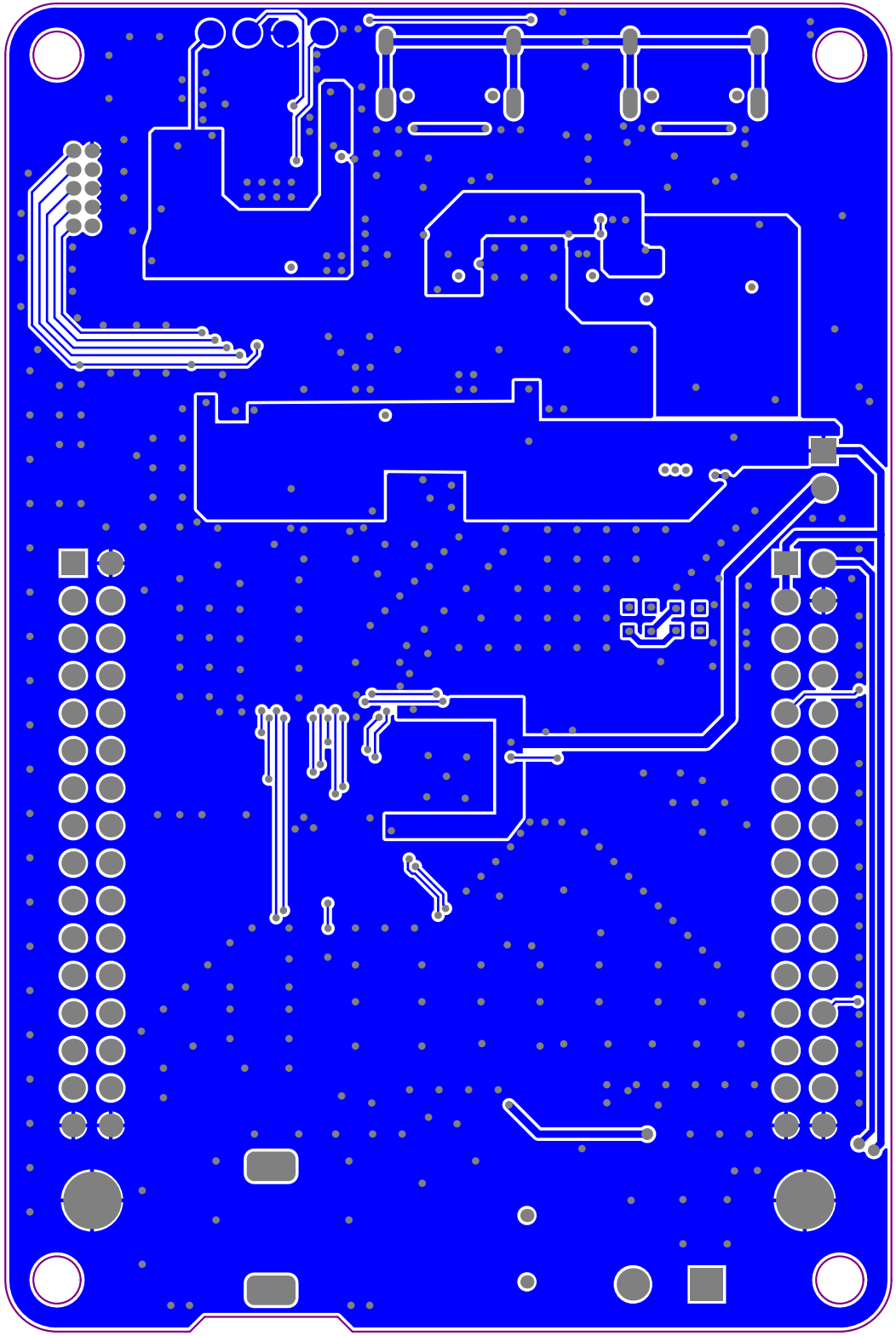


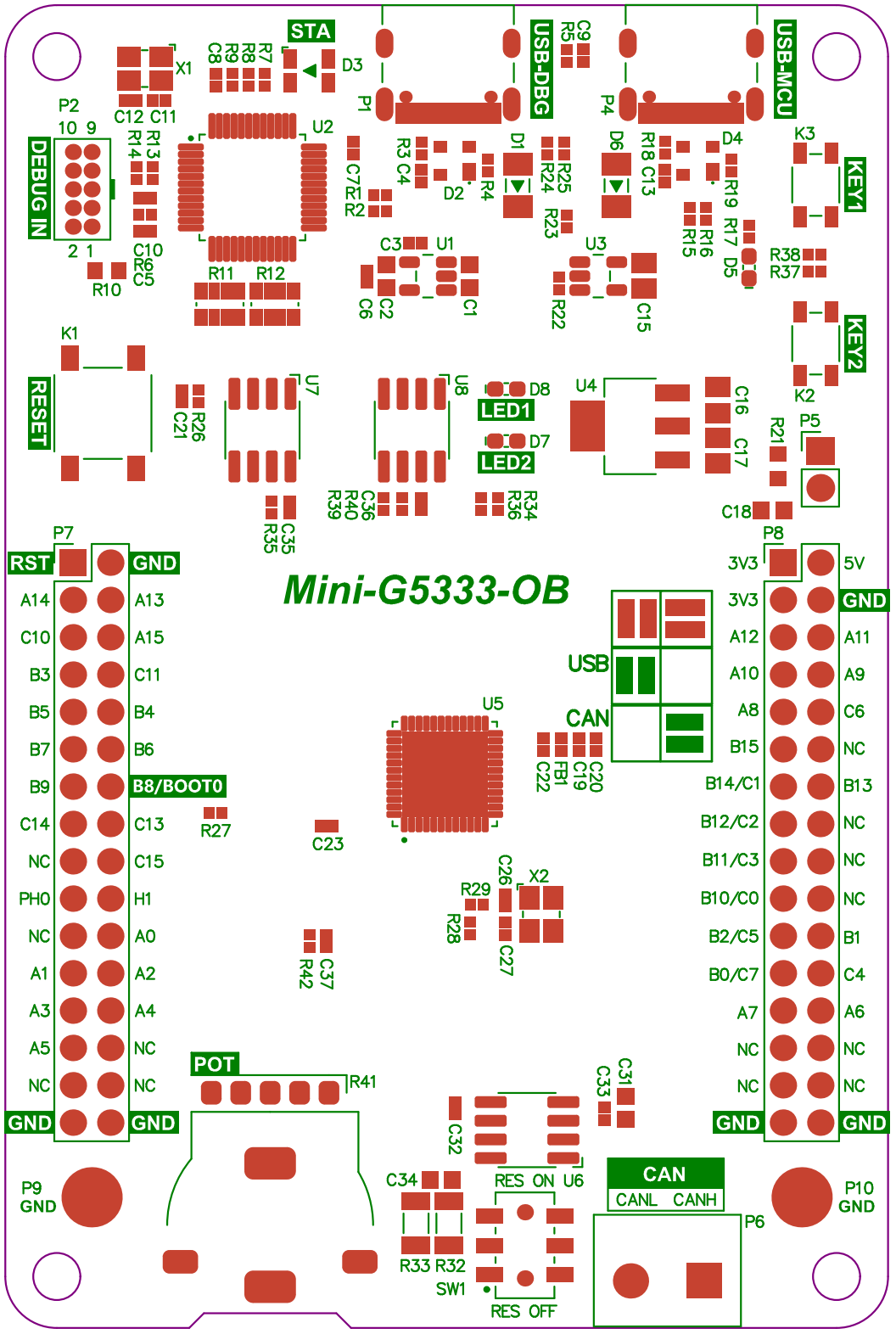
DNP : DO NOT POPULATE

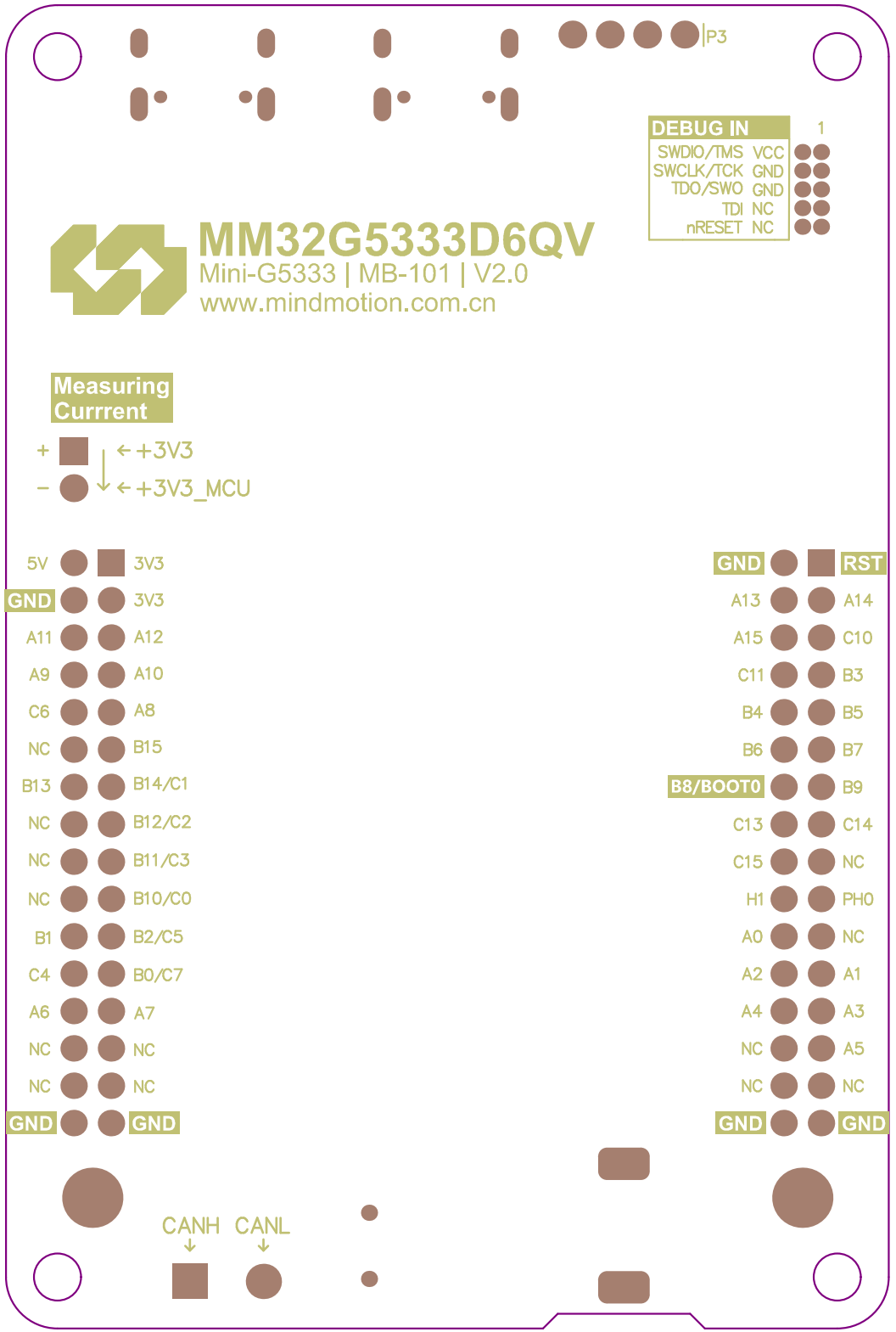
Project Title	MB101-Mini-G5333-OB		
Sheet Title	06---Onboard Source		
Size A3	Rev A	Drawn	Lu Weiwei
Sheet 6 / 9		Checked	Li Hongpeng
Data 2023.07.03		Approved	Chen Do

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Mini-G5333-OB
STA
USB-DBG
USB-MCU
DEBUG IN
RESET
KEY1
KEY2
LED1
LED2
USB
CAN
POT
RST
GND
B8/BOOT0
3V3
5V
RES ON
RES OFF
CANL CANH
P1
P2
P4
P5
P6
P7
P8
P9
P10
U1
U2
U3
U4
U5
U6
U7
U8
X1
X2
D1
D2
D3
D4
D5
D6
D7
D8
K1
K2
K3
SW1
R41
C23
C31
C32
C33
C34
R33
R32


P3
DEBUG IN
1
SWDIO/TMS VCC
SWCLK/TCK GND
TDO/SWO GND
TDI NC
nRESET NC
MM32G5333D6QV
Mini-G5333 | MB-101 | V2.0
www.mindmotion.com.cn
Measuring Currrent
+
−
←+3V3
←+3V3_MCU
5V 3V3
GND 3V3
A11 A12
A9 A10
C6 A8
NC B15
B13 B14/C1
NC B12/C2
NC B11/C3
NC B10/C0
B1 B2/C5
C4 B0/C7
A6 A7
NC NC
NC NC
GND GND
GND RST
A13 A14
A15 C10
C11 B3
B4 B5
B6 B7
B8/BOOT0 B9
C13 C14
C15 NC
H1 PH0
A0 NC
A2 A1
A4 A3
NC A5
NC NC
GND GND
CANH CANL